

Fundamental MATLAB Programming for AI

Today's goals

Module 1
The Fundamentals of MATLAB

Module 2
Produce Effective MATLAB Programs for Science and Engineering Problems

Module 3
Get Ready to Learn AI using MATLAB

Module 3 Introduction

Introduction to Module 3's Study Video • 1 min

Module 3-1 MATLAB Table and Cell Arrays

MATLAB Table and Cell Arrays: MATLAB Table Arrays Video • 10 min

MATLAB Table and Cell Arrays: Import/Export Tables Video • 4 min

MATLAB Table and Cell Arrays: MATLAB Cell Arrays Video • 9 min

Materials about MATLAB Table and Cell Arrays Reading • 10 min

Module 3-2 Categorical Data

Categorical Data: Categorical Data and One Hot Encoding Video • 12 min

Materials about Categorical Data Reading • 10 min

Module 3-3 Write an Objective Function

Write an Objective Function: MSE and Rastrigin Benchmark Explained Video • 7 min

Write an Objective Function: Understanding and Implementing Binary Cross Entropy Video • 4 min

Materials about Objective Functions Reading • 10 min

Module 3-4 Data Cleaning

Data Cleaning: Essentials Video • 3 min

Materials about Data Cleaning Reading • 10 min

Preparing Reliable Data and Objectives for AI in MATLAB Dialogue • 5 min

Module 3 Quiz

Module 3 Quiz Practice Assignment • Grade: 100%

Module 3 Practice/Warm-up Exercise for Graded Assignment

Module 3 Exercise 1 Ungraded App Item • 30 min

CSV file in Exercise 1 Reading • 1 min

How to Use MATLAB Online for Module 3 Exercise 1 Video • 6 min

Exercise Solutions Reading • 10 min

Module 3 Assignment

Module 3 Assignment 1 Graded App Item • Grade: 100%

CSV file for Assignment 1 Reading • 1 min

Module 3 Assignment 2 Graded App Item • 30 min

CSV files for Assignment 2 Reading • 1 min

Module 3 Recap Reading • 5 min

Keeping Learning with UofG Online Reading • 5 min

Module 3 Recap

0:03 / 1:02

2x

In this module, you learned how to:

- Work with structured data using tables, cell arrays, and categorical variables
- Import, organize, and transform datasets for analysis
- Define and implement objective functions such as MSE and binary cross-entropy
- Identify and correct data issues, including duplicates, missing values, and inconsistent scales
- Perform grouped analysis and compute summary statistics (mean, standard deviation)
- Apply logical indexing and structured workflows to analyze real datasets

At this stage, you should be able to:

- Choose appropriate data structures based on the problem and workflow
- Clean datasets while understanding the assumptions behind each operation
- Validate every step of your data processing (before and after changes)
- Detect common sources of error such as hidden duplicates, NaNs, and incorrect indexing
- Interpret statistical results with awareness of variability, not just averages
- Translate raw data into structured inputs suitable for AI and optimization tasks.

Go to next item

✓ Completed

Like

Dislike

Report an issue